

BANARAS HINDU UNIVERSITY

B.Com. (Hons.) — Semester IV Examination, 2023-24

Paper No. BCH-224: Business Mathematics

Time: 3 Hours

Maximum Marks: 70

*Instructions: Attempt all questions. All questions carry equal marks (14 marks each).***Question 1.**

- (i) The p th term of an A.P. is q and q th term is p , prove that m th term will be $p + q - m$.
[7]
- (ii) Sum of 8 terms of an A.P. is 64 and sum of 16 terms is 256. Find the sum of n terms.
[7]

OR

- (i) 7th and 13th terms of an A.P. are 13 and 25 respectively. Find the sum of 10 terms. [7]
- (ii) Sum of 3 terms of an A.P. is 15 and their product is 105. Find the numbers. [7]

Question 2.

- (i) If the sum of n terms of a G.P. $\frac{2}{9} - \frac{1}{3} + \frac{1}{2} - \dots$ is $-\frac{1261}{576}$, find n . [7]
- (ii) A person desires to save Rs. 2 in January, Rs. 4 in February, Rs. 8 in March, Rs. 16 in April and so on upto the end of the year. Determine:
- (a) How much amount he will save during one year? [3.5]
- (b) What will he save in the month of October? [3.5]

OR

- (i) $(p + q)$ th term of a G.P. is m and $(p - q)$ th term is n . Find p th term. [7]
- (ii) Arithmetic mean between two numbers is 10 and their geometric mean is 8. Find the numbers. [7]

Question 3.

- (i) If ${}^{2n+1}P_{n-1} : {}^{2n-1}P_n = 3 : 5$, find n . [7]
- (ii) A box contains 7 red, 6 white and 4 blue balls. How many selections of 3 balls can be made so that:
- (a) None is red. [3.5]
- (b) One is of each colour. [3.5]

OR

(i) Prove that: ${}^{n-1}P_r + r \cdot {}^{n-1}P_{r-1} = {}^nP_r$. [7]

(ii) If ${}^{2n}C_3 : {}^nC_2 = 44 : 3$, find n . [7]

Question 4.

(i) Find the term independent of x in the expansion of $\left(x^2 + \frac{1}{x}\right)^{12}$. [7]

(ii) If in the expansion of $(1+x)^{24}$ the coefficient of r th term and $(r+4)$ th term are equal, find r . [7]

OR

Prove that the middle term in the expansion of $(1+x)^{2n}$ is $\frac{1 \cdot 3 \cdot 5 \dots (2n-1)}{n!} 2^n x^n$. [14]

Question 5.

(i) If $f(x) = \frac{x-1}{x+1}$, then find the value of $\frac{f(x)-f(y)}{1+f(x)f(y)}$. [4]

(ii) Evaluate: $\lim_{x \rightarrow \infty} \frac{3x^2+5x-5}{5x^2+6x+2}$. [4]

(iii) Differentiate w.r.t. x : $\frac{(x+1)^2 \sqrt{x-1}}{(x+4)^3 e^x}$. [6]

OR

(i) Simplify: $\int \frac{\sqrt{x+3}\sqrt{x}}{5x\sqrt{x}} dx$. [4]

(ii) Simplify: $\int \frac{1}{\sqrt{x+3}-\sqrt{x}} dx$. [4]

(iii) Simplify: $\int \frac{(x+1)e^x}{(x+2)^2} dx$. [6]